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# The benefits for European companies and investors of promoting happiness through high-performing work policies

Investors of  
promoting  
happiness  
through HPWP

Oscar Valdemar De la Torre-Torres

*Facultad de Contaduría y Ciencias Administrativas,  
Universidad Michoacana de San Nicolás de Hidalgo, Morelia, Mexico*

María Isabel Martínez Torre-Enciso

*Departamento de Financiación e Investigación Comercial,  
Facultad de Ciencias Económicas y Empresariales,  
Universidad Autónoma de Madrid, Madrid, Spain*

María de la Cruz Del Río-Rama

*Business Management and Marketing, Universidade de Vigo, Vigo, Spain, and*

José Álvarez-García

*Facultad de Empresa Finanzas y Turismo, Departamento de Economía Financiera y  
Contabilidad, Universidad de Extremadura, Cáceres, Spain*

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## Abstract

**Purpose** – In this paper, the authors tested if promoting the workforce's happiness (through high performance work policies or HPWP) and well-being in European Public companies relates to their profitability (return on equity, ROE), market risk (beta) and stock price return. Also, the authors tested if investors have a performance benefit if they buy a portfolio screened with companies with HPWP.

**Design/methodology/approach** – The authors proxied the quality of the HPWP efforts in the first method with the Refinitiv workforce score. They used this data in an unbalanced panel of eastern, western, northern and southern Europe companies from 2011 to 2022. The panel data also included the ROE, the market risk (beta) and the stock price return of these companies. The authors estimated the corresponding regressions with the panel data and tested the relationship between the workforce score and these three variables. In a second method, they simulated the weekly performance of a portfolio that invested only in European companies with high standards in their HPWP and compared its performance against a conventional market portfolio (with no HPWP screening).

**Findings** – In the first method, the authors found no significant relationship between the workforce score and the ROE, beta, or stock price return in the panel regression, controlling for random effects. In the second one, they found no over or underperformance in the HPWP portfolio against the European market one in the second method.

**Practical implications** – The results suggest that there is no risk or cost for European Public companies and investors alike if they promote, with better HPWP, the happiness and well-being of their workforce. The findings suggest that if European companies promote HPWP, there will be no adverse impact on their profits, market risk, or stock price performance. Also, investors will not lose performance (against a conventional market portfolio) if they screen their portfolios with this type of workforce-friendly companies.

**Originality/value** – Increase the scarce literature on the test of the workforce score with company profitability (ROE), stock market price variation and stock market risk level.

**Keywords** Workforce happiness, Workforce well-being, High-performing working practices, Company performance, Profitability, Stock price performance

**Paper type** Research paper



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## 1. Introduction

The influence of the workforce's happiness on company performance has been a topic of interest since the beginning of management sciences. The general assumption is if companies promote policies to enhance their workforce happiness, health and group integration, they will be more profitable. The rationale is that the proper motivation of the workforce leads to more production (more profits, more value to shareholders and less market risk exposure). Departing from this rationale, companies could engage in practices for their workforce's well-being. The policies that promote happiness, health and group integration are known as high performance work policies (HPWP).

Also, the well-being of the workforce is a multidimensional concept that results from three dimensions ([Grant et al., 2007](#); [Misselbrook, 2014](#)): Happiness (psychological well-being, a dimension of interest herein), Health (physical well-being) and Social relationship (social well-being). The first of these three has attracted attention in management because happy people tend to be more productive.

Several works have tested the cause-effect of happiness (or well-being) and the production level. This relationship is one of the most exciting and fundamental in labor economics ([Veenhoven, 1988](#)). This internal company performance perspective is one that we want to test in European companies traded in the leading European stock exchanges that are members of the Refinitiv Europe price index. An index of the 1,828 most prominent and traded companies in Eastern, Northern, Southern and Western Europe. More specifically, this paper's authors want to test this relationship from four perspectives:

- (1) The benefit of the internal relation between performance (company profits or return on equity, ROE) and the HPWP in the company,
- (2) The benefit of HPWP quality in stock price performance (price percentage yearly variation),
- (3) The corresponding benefit of HPWP quality in stock market company's risk proxied with the capital asset pricing model ([Sharpe, 1963, 1964](#)) or CAPM's beta ( $\beta$ ), and
- (4) The answer to the following questions: Is it beneficial for a given investor to promote activist investing by managing a portfolio invested (positively screened) in high HPWP quality companies? Will she or he have a better or at least similar risk-return relationship with this HPWP portfolio against a broad-market capitalization one?

The quality of HPWP in a company is a subject that is also of interest in the context of corporate social responsibility (CSR) and in environmental, social and governance (ESG) investing. ESG investing is an emerging investment style that focuses on a screening process of companies with high ESG practices ([Derwall et al., 2011](#)). It is an investment style that promotes investment only in companies with high ESG (or CSR) standards. In the social dimension of ESG, the quality or well-being of the workforce is essential. Therefore, the proper scoring process of a company's workforce is part of the general fully integrated ESG score that third-party companies (such as Eiris-MSCI, Robeco, Sustainalytics, or Refinitiv) quantify for several public companies around the world. By following an internal and financial report review, the opinion in social media, news, or with data of public organisms or non-governmental organizations (NGO), these companies measure several items. Items that lead to an ESG, social, environmental, or governance company score each year, given a proper scoring methodology. This score has values from zero to 100, the highest ESG level. As mentioned, from the three dimensions, the score of interest is the workforce score in the social dimension in the [Refinitiv \(2019\)](#) ESG methodology. This score, given 64 items related to the execution of HPWP actions or qualities, measures the level at which the company of interest promotes happiness, health, social relationship and workers' personal development and inclusion.

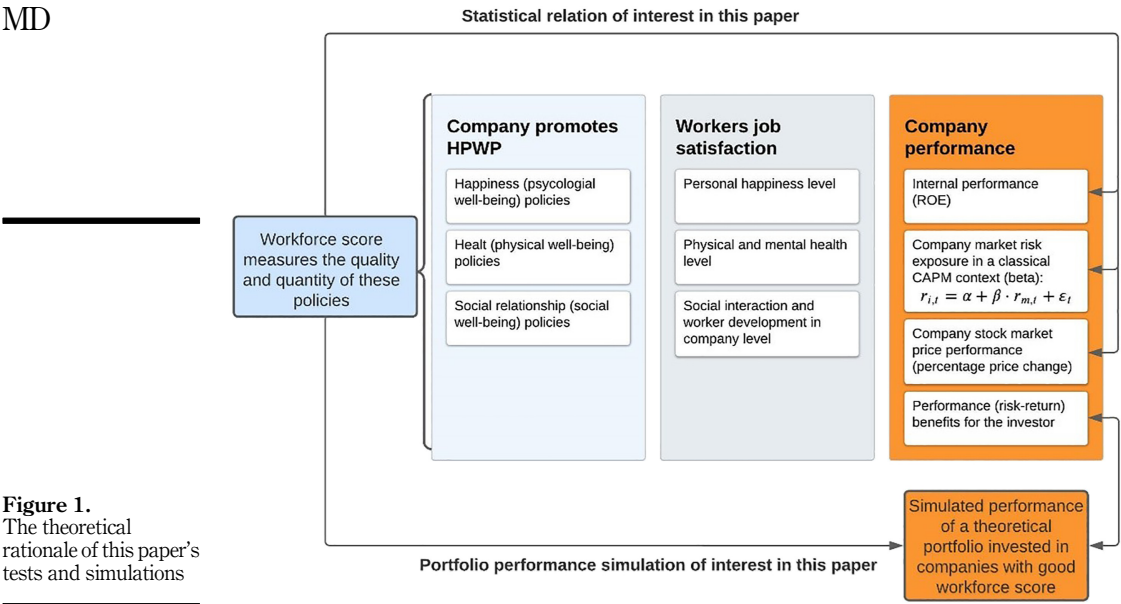
At this point, it is interesting to highlight the close theoretical and practical relationship that workforce happiness, CSR and ESG (as a social responsibility framework) investment have. At first sight, these could look like unrelated concepts or terms. Workforce happiness and workforce well-being are terms in the realm of management theories and some aspects of labor economics. At first, these refer to the benefit of a productive workforce due to proper psychological (or happiness, one of the main interests in this journal), social, or physiological health motivation. On the other hand, CSR practices and standards of a company are terms in management sciences and business philosophy. The core discussion relates to the proper company practices and their benefits in environmental, social, or governance activities. This theoretical term is widely discussed in management sciences (a discussion we do not follow herein). It focuses on the role and impacts that a company must have in these three dimensions of interest. ESG is a more finance industry-specific term. The first mention of these three dimensions is in the [United Nations \(UN\) Environmental Programme \(2004\)](#) report. A report with recommendations by the UN and a group of financial institutions to integrate a company's CSR quality in the investment process. This report led to the development of a practice known as socially responsible investment (CSR) or ESG investing. With this report mentioned above, the following UN Millennium declaration ([United Nations, 2005](#)), and the UN's principles for responsible investment ([United Nations, 2006](#)), ESG emerged as a CSR scoring framework and ESG investing as a widespread practice among institutional and private investors.

As part of the social pillar of the ESG framework, the quality of workforce management and the resulting workforce's well-being (and happiness) is an essential dimension for the company's (and investor's) performance. The primary rationale is: if a company promotes good workforce management (high-performing working policies), its workforce's well-being (and happiness) will increase. If this result holds, it is expected that the company will have higher productivity and profitability. Consequently, the company's stock market price should reflect this positive relationship with a higher level and a lower market risk exposure (one of the main results to test herein). As a result of the theoretical relationship between ESG investing, CSR and HPWP; HPWP relates to a better, happier and healthier workforce. This practice is part of CSR practices that, in the ESG scoring context, are part of the social pillar. If a company promotes HPWP, it promotes a better social pillar score and, consequently, its profitability and stock price performance.

Departing from the previous theoretical and practical motivations, [Figure 1](#) presents this paper's theoretical framework. Given the literature review, the authors of the present paper found that there is scant literature about the test of the workforce score with company profitability (ROE), stock market price variation and stock market risk level. As mentioned earlier, there are proofs of the benefit of implementing HPWP in workforce happiness and well-being. Departing from these results, the present paper's authors accept as given this relation and focus only on the HPWP quality-company performance one. The authors extend the current tests to the European stock market companies for the ROE, market risk exposure (beta) and stock price percentage change. They did this to extend the review of this relationship in these stock markets.

The theoretical background of the test is to expand the current results about labor economics and ESG investing. By assuming, as given (with the previous literature review), that high HPWP leads to workers' happiness and this result to better company performance, the authors tested the following three hypotheses:

- H1.* "The higher the quality of HPWP, proxied with the workforce score, the higher the profitability (ROE) in a given company."
- H2.* "The higher the quality of HPWP, proxied with the workforce score, the lower the company's market risk exposure, proxied with the CAPM beta."
- H3.* "The higher the quality of HPWP, proxied with the workforce score, the better the company's yearly stock price percentage variation."



**Figure 1.**  
The theoretical rationale of this paper's tests and simulations

The first perspective has a Microeconomics one in terms of labor economics. It tests the validity of this research area by proving that happier workers lead to a more profitable company. The authors' position is against the conventional classical Economics that holds that investing in HPWP reduces profits, proxied with the ROE.

The second and third hypotheses have a macroeconomics perspective because they model the aggregated behavior in European stock markets and assume that stock prices reflect all the information about company profits, performance and particular aspects of its management. [Krausert \(2016\)](#) extended [Akerlof's \(1978\)](#) need for signals to enhance informational efficiency in stock markets and suggested a potentially positive and significant relationship between the disclosure of HPWP and stock performance. In the case of this research paper, the HPWP signal is the third-party (Refinitiv) workforce score. The higher the score, the better the signal and the better the performance of the European company. Therefore, we extend Akerlof and Krausert's reviews to the European stock markets.

Finally, to enhance the benefit for stockholders. That is, the value transmission of the quality of HPWP to profits and, then, to investors, the authors of this paper simulated the weekly performance of a high-quality workforce score-screened portfolio. The authors' fourth hypothesis was:

*H4.* "The performance of a portfolio invested in HWPW (high workforce score) stocks is better or, at least, equal to that of a market diversified portfolio."

In this paper, the authors approximated the level of workers' well-being to the quality and quantity of HPWP. The higher the quality and quantity of HPWP, the higher the ESG score of a company's workforce. Based on this reasoning, the authors used Refinitiv's ESG scoring method. More specifically, they used the workforce score to test the above hypotheses. For a detailed introduction to the Refinitiv ESG scoring method and the estimation of the workforce score, please refer to [Appendix](#).

With this paper's results, it is expected that European Academics, practitioners in ESG investing and the investment industry could feel compelled to promote workforce happiness and well-being. This result will hold if they invest in companies with high quality in their HPWP. Related to this result and the second hypothesis, the authors expect that if the market risk exposure is lower (due to the shunned-stock hypothesis of [Derwall \*et al.\* \(2011\)](#)), the company's stock demand rises due to its HPWP.

Given the goals of the present paper, along with the theoretical background of the tests made, the next section will present a brief literature review on the topic. The third section describes how was the input data processed for the tests. In the fourth section, the authors present the main results and findings, and finally, in the fifth section, the paper concludes and gives the guidelines for further research.

## 2. Literature review

Several works have tested the cause-effect of happiness (or well-being) and the production level. This relationship is one of the most exciting and fundamental in labor economics ([Veenhoven, 1988](#)). One of the first works testing this relation is the one of [Wright \*et al.\* \(2007\)](#). In this paper, the author reviewed the influence of job satisfaction and psychological well-being (happiness) in supervisory performance ratings, finding a positive effect in the former over the latter. [Oswald \*et al.\* \(2015\)](#) made a conductist test on students of top universities in England. They found that a person receiving a positive stimulus (such as the view of a comic movie) tends to perform better in an enrollment test. Also, if a person feels bad for a personal loss, the opposite happens in their productivity. [Magnier-Watanabe \*et al.\* \(2017\)](#), [Chia-Hao and Ting-Ya \(2018\)](#) and [Pradenas \*et al.\* \(2021\)](#) made similar tests in Japan, Taiwan and Chile. They found similar results on the influence of company performance from happiness or a more general form of well-being. From another perspective, [Popescu \*et al.\* \(2022\)](#) measured the mutual effect of employee engagement (implementation of some states of HPWP) and well-being among workers of Rumania. The authors found a positive and bijective relationship among these two variables, highlighting the importance of company actions to enhance workers' happiness or (a more general concept in labor economics) well-being. These previous works prove that there is a positive effect of HPWP on workers' well-being.

Following this line of research (the influence of HPWP), the works of [Ravina-Ripoll \*et al.\* \(2021\)](#), [Yoshida \*et al.\* \(2021\)](#), [Rodríguez \*et al.\* \(2020\)](#) made a review of the external loyalty of consumers to companies that tested the positive relationship between consumers loyalty programs and the corresponding client's loyalty level. Following this rationale, [Momparrer \*et al.\* \(2011\)](#) measured how contribution, conviction and culture (feeling about the current job), commitment and confidence contribute to work happiness. [Sattar \*et al.\* \(2015\)](#), partly related to the previous work, measured the influence of HPWP on job satisfaction and the benefit of the former on job productivity. This conclusion is similar to the works of [Ahmed \*et al.\* \(2020\)](#) and [Muthuveelo \*et al.\* \(2022\)](#).

As noted in this brief literature review, a positive relationship exists between HPWP and workers' happiness or (a broader concept) worker's well-being. Also, a positive relationship exists between workers' well-being (the more general concept used herein) and company productivity.

The literature that relates the level of ESG or pillar-specific scores with company profits, stock price performance and market risk is broad. [Brogi and Lagassio \(2019\)](#), [Xie \*et al.\* \(2019\)](#), [Sethi \*et al.\* \(2017\)](#) and [Crifo \*et al.\* \(2016\)](#) are some recent examples. Also, the literature that favors or refutes the better performance of ESG companies' stock price against conventional (not so ESG) or sinful (with high ESG negative impact) ones is broad. Despite this, the present paper's authors only found the works of [Edmans \(2012\)](#), [Li and Zhang \(2013\)](#), [Cao and Chen \(2016\)](#), [Flammer and Bansal \(2017\)](#) and [Goetzel \*et al.\* \(2019\)](#) in the benefit of HPWP in stock price. A gap that the authors pretend to fill in this paper.

The previous literature review (among other works the authors omitted due to space constraints) gives a theoretical background to the tests in the first hypothesis. The papers of [Edmans \(2012\)](#), [Li and Zhang \(2013\)](#), [Cao and Chen \(2016\)](#), [Flammer and Bansal \(2017\)](#) and [Goetzel \*et al.\* \(2019\)](#) are among the first works that tested and give theoretical background to the second question tested in this paper (the stock price performance due workers' well-being and their motivating HPWP). These authors used two perspectives to test this relationship. First, they estimated a multifactor model (such as the [Carhart \(1997\)](#) one) of US or UK stocks of companies members of the Fortune "best place to work" list. They found a significant "alpha" or inertial return (the intercept of either panel or robust cross-sectional regression) in their historical performance. From the other perspective, the authors made groups of US companies with high internal health (well-being) promotion policies and external (clients or external stakeholders) ones and compared the performance against the broader S&P 500 stock index. The results, in line with the previous theoretical perspective, suggest a better stock performance in companies with better HPWP. These five works (among other related non-quoted herein for space purposes) motivate this work's second and fourth perspectives. Little or nothing is written for the European stock markets as a group or portfolio (excluding the UK or Germany). Therefore two questions emerge: (1) Will the HPWP positively influence stock price? and (2) Does an HPWP portfolio (similar to the one of [Goetzel \*et al.\* \(2019\)](#)) lead to better (or at least equal) performance than the Refinitiv Europe price return index? In this paper, a panel regression model tested the influence between the workforce score (an ESG-score that measures the quality of HPWP) and annual percentage price variation. Also, the authors simulated the performance of a hypothetical portfolio invested in companies with HPWP practices at an investment level related to these. That is, this hypothetical portfolio invested only in companies that reported an HPWP (working force) score from Refinitiv, and the investment level is higher in companies with a higher score.

The third perspective (the influence of HPWP quality on company market risk) is motivated by the works of [Cao and Chen \(2016\)](#), [Callen and Fang \(2015\)](#) and [Ben-Nasr and Ghouma \(2018\)](#). These works measured either the performance that high HPWP quality companies had in distress market periods or the probability of a price crash (significant price loss) against other non-HPWP companies. These works, motivated by the stakeholder's or the agency's theories, found evidence that high levels of employee well-being contribute to stock price crash risk because these enhance employee loyalty and reduce the likelihood of speaking about internal performance bad news from the employees. Employees "tend to quiet bad news due to more commitment to the company."

Given this brief literature review, the paper proceeds with the test of the four hypotheses of interest.

### 3. Data gathering and processing

To test the first three hypotheses, the yearly  $ROE_{i,t}$ , yearly percentage price variation and the weekly three-year weighted average cost of capital's (WACC) beta ( $\beta_{i,t}$ ) were downloaded from the databases of Refinitiv. As an investment universe, the authors fetched this yearly data from 2010 to 2021 (the most recent ESG yearly score values at the moment of writing this paper) for all the stock members of the Refinitiv Europe price return index (with the TRXFLDEUPU Refinitiv Identifier Code or RIC) as of August 28th, 2022. This fetching process led to the first screen of 1,828 stock from Eastern, Northern, Southern and Western Europe. For the first three hypotheses, we used a second filter on the companies that reported, at least one year, the ROE, beta, stock price percentage variation, ESG score, social pillar score, governance score, and, most important, the workforce score. If a given year in a given company did not report one of these scores, that company's (year) row was omitted. This filter led to an unbalanced panel of 9,325 observations with 1,537 companies and a 1–16 years' time

series length. To test the first three hypotheses, we estimated the following panel regression equations with the pooling, fixed-effect (within groups) and random effects:

$$H1 : \widehat{ROE}_{i,t} = \alpha + \beta_1 ESG_{i,t} + \beta_2 ENV_{i,t} + \beta_3 SOC_{i,t} + \beta_4 GOV_{i,t} + \beta_5 WorkHap_{i,t} \quad (1)$$

$$H2 : \widehat{r}_{i,t} = \alpha + \beta_1 ESG_{i,t} + \beta_2 ENV_{i,t} + \beta_3 SOC_{i,t} + \beta_4 GOV_{i,t} + \beta_5 WorkHap_{i,t} \quad (2)$$

$$H3 : \widehat{\beta}_{i,t} = \alpha + \beta_1 ESG_{i,t} + \beta_2 ENV_{i,t} + \beta_3 SOC_{i,t} + \beta_4 GOV_{i,t} + \beta_5 WorkHap_{i,t} \quad (3)$$

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In the previous expressions  $\widehat{ROE}_{i,t}$  is the  $i$ th company ROE at year or time  $t$ .  $\widehat{r}_{i,t}$  is that years/ company stock price percentage variation or return.  $\widehat{\beta}_{i,t}$  is the weekly three-year CAPM beta estimated by Refinitiv.  $ESG_{i,t}$ ,  $ENV_{i,t}$ ,  $GOV_{i,t}$  and  $SOC_{i,t}$  are the ESG pillar scores, respectively.  $WorkHap_{i,t}$  is the worker's happiness or well-being score. That is the Refinitiv workforce score.

We estimated these regressions in four steps or four different functional forms. The first one with the  $ESG_{i,t}$  only, the second with the  $ENV_{i,t}$ ,  $SOC_{i,t}$  and  $GOV_{i,t}$  scores (the disaggregated pillar-specific ESG score), the third with  $ENV_{i,t}$ ,  $GOV_{i,t}$  and  $WorkHap_{i,t}$  and a fourth model with only  $WorkHap_{i,t}$ . We estimated these four regression sequences to test once again the relation between ESG practices and company performance relation to test if this positive relationship holds in our investment universe and to test if the relationship holds from the general ESG framework to the particular  $WorkHap_{i,t}$  context. In the fourth and fifth regressions  $SOC_{i,t}$  was omitted because we avoided potential multicollinearity issues. We used the Newey-West (1986) robust standard error estimation method to reduce the potential impact of heteroskedasticity and serial correlation. Because the ESG and workforce scores are estimated ex-post (the result of year  $t$  corresponds to the ESG grading of year  $t - 1$ ), we included the one-year lag of the five regressors in (1), (2) and (3) to capture the signal effect of the ESG score perception at  $t$ , given the published data from  $t - 1$ .

To test hypotheses H1 to H3,  $\beta_5$  must be significant and positive in H1 and H3 and significant and negative in H2.

To test the fourth hypothesis, we simulated the performance of a hypothetical portfolio that invested in companies with a high workforce score. To measure this strategy, we fetched the weekly Euro-based market capitalization data of the stock members of the TRXFLDEUPU index. With the market capitalization  $MP_{i,t}$  of a given company at  $t$ , we calculated a first investment level  $\psi_{i,t}$  as follows:

$$\psi_{i,t} = \frac{MP_{i,t}}{\sum_{i=1}^N MP_{i,t}} \quad (4)$$

With the work happiness or workforce score ( $WorkHap_{i,t}$ ) we estimated a capped investment level (ceiling or maximum investment level) by normalizing the reported  $WorkHap_{i,t}$  score:

$$ceiling_{i,t} = \frac{WorkHap_{i,t}}{\sum_i WorkHap_{i,t}} \quad (5)$$

Once we estimated the ceiling or maximum investment level in each company, we multiplied that ceiling by the investment level  $\psi_{i,t}$  in (4) with the  $i$ th corresponding ceiling level in (5). This calculation led to the final investment level of each stock in the simulated portfolio:

$$\omega_{i,t} = \psi_{i,t} \cdot ceiling_{i,t} \quad (6)$$

If a given company has no  $WorkHap_{i,t}$  score at  $t$ , it is expected that the value of  $ceiling_{i,t}$  will be zero. Therefore the simulated portfolio will not invest in that company ( $\omega_{i,t} = 0$ ). In the present simulations, this simulated portfolio was the market-capitalization workforce happiness-capped portfolio (MktWorkforceCap).

We also simulated a second theoretical portfolio in which the investment levels are *ceiling* in (5). This weighting method uses fundamental or company data for this purpose, and this paper's authors named this portfolio the fundamental workforce portfolio (FundamentalWorkForce). Similarly, the authors used the market capitalization weighting method in (4) to simulate the European market (MktPortfolio).

For the simulations, the authors also fetched the historical price ( $P_{i,t}$ ) of each stock member of the TRXFLDEUPU index from Refinitiv and calculated the continuous-time return as follows:

$$r_{i,t} = \ln(P_{i,t}) - \ln(P_{i,t-1}) \quad (7)$$

With this continuous-time return in (7), they estimated the simulated portfolio return with the investment level in (6):

$$r_{p,t} = \omega_{i,t} \cdot r_{i,t} \quad (8)$$

For the case of the MktPortfolio, we estimated its simulated return by using the market capitalization weighting method as in (4):

$$r_{m,t} = \sum_i^n \psi_i \cdot r_{i,t} \quad (9)$$

The historical performance of the market portfolio  $r_{m,t}$  was a proxy of the entire European market performance in the investment universe.

To test the fourth hypothesis, we compared the accumulated return of the work happiness portfolio ( $r_{p,t}$ ) and compared its historical performance with the market portfolio with the observed returns of both portfolios and their Sharpe (1966) ratios:

$$SR_{p,t} = \frac{r_{p,t} - rf}{\sigma_{p,t}} \quad (10)$$

$$SR_{m,t} = \frac{r_{m,t} - rf}{\sigma_{m,t}} \quad (11)$$

For the Sharpe ratios, we used the three-month bundes German bonds rate as the risk-free rate ( $rf$ ) and the 52 weeks (one year) returns to estimate the portfolio's standard deviation ( $\sigma_{p,t}$ ,  $\sigma_{m,t}$ ) of (10) and (11). We used the classical portfolio Math of Markowitz (1952, 1959) to estimate the simulated portfolios' variance. With the use of a one-way ANOVA and Kruskal–Wallis tests, we compared the statistical equality of the returns ( $r_{p,t}$  and  $r_{m,t}$  values) and Sharpe ratios ( $SR_{p,t}$ ,  $SR_{m,t}$ ) simulated values. To prove the fourth hypothesis this must happen in the ANOVA (and Kruskal–Wallis) test:

- (1) The  $p$ -values must be lower than 5.00% (the significance threshold used herein), and a visual box-plot with higher values returns or Sharpes in MktWorkforceCap or FundamentalWorkForce against the MktPortfolio, or
- (2) The ANOVA tests  $p$ -values must be higher than 5.00%, suggesting statistically equal values.

We also estimated the CAPM auxiliary regression with the simulated portfolios' as follows:

$$r_{p,t} = \alpha + \beta_m r_{m,t} + \varepsilon_t \quad (12)$$

To test the fourth hypothesis,  $\beta_m \geq 1$  in (12). This conclusion will indicate that the performance of the market and the simulated portfolios have statistically equal performance. These results will evidence the statistically equal performance of the simulated portfolio with the market one. We didn't make and include the cointegration test in the paper.

To estimate the simulated values ( $P_{p,t}$ ,  $P_{m,t}$ ) of the simulated and market portfolio, we used the corresponding portfolio returns in (8) and (9) to estimate the January 1st, 2005, 100 value of each portfolio.

Also, with the use of (12) in each portfolio, we estimated Jensen's (1968) alpha and the corresponding Treynor and Black (1973) ratio to determine if there is a risk premium of the simulated portfolios against the MktPortfolio. Similarly, we used the Hirschman (1964) ratio to test the concentration level in the simulated portfolios.

The authors present the list of the European companies used in the simulations and the panel data regression in the supplementary material < supplementary-material-link-lies-herein-if-the-paper-is-accepted >.

Once the authors detailed how they processed the input data and tested the four working hypotheses, they will discuss the results and findings of the tests and simulations.

#### 4. Simulations and hypotheses tests results

For the case of the first three hypotheses, the authors estimated the three types of panel data regression (pooled regression, fixed-effects within groups and random effects) and determined, for the five previously mentioned regressions in (1), (2) and (3), the  $F$  and Hausman (1978) tests. Once they determined the best-fitting panel data regression model and as mentioned previously, they estimated the subsequent four regressions for the three dependent variables of interest ( $ROE_{i,t}$ ,  $\beta_{i,t}$ ,  $r_{i,t}$ ):

$$\text{dependent variable} = \alpha + \beta_1 ESG_{i,t} + \beta_2 ESG_{i,t-1} + \varepsilon_{i,t} \quad (13)$$

$$\begin{aligned} \text{dependent variable} = & \alpha + \beta_1 ENV_{i,t} + \beta_2 ENV_{i,t-1} + \beta_3 SOC_{i,t} + \beta_4 SOC_{i,t-1} + \beta_5 GOV_{i,t} \\ & + \beta_6 GOV_{i,t-1} + \varepsilon_{i,t} \end{aligned} \quad (14)$$

$$\begin{aligned} \text{dependent variable} = & \alpha + \beta_1 ENV_{i,t} + \beta_2 ENV_{i,t-1} + \beta_3 SOC_{i,t} + \beta_3 SOC_{i,t-1} + \beta_7 GOV_{i,t} \\ & + \beta_8 GOV_{i,t-1} + \varepsilon_{i,t} \end{aligned} \quad (15)$$

$$\text{dependent variable} = \alpha + \beta_7 GOV_{i,t} + \beta_8 GOV_{i,t-1} + \varepsilon_{i,t} \quad (16)$$

To prove as true the first three hypotheses, the values of  $\beta_7$  and, especially,  $\beta_8$  must be significant and according to their expected sign.

**4.1 H1:** “The higher the quality of HPWP, proxied with the workforce score, the higher the profitability (ROE) in a given company”

In Table 1, the authors present the  $p$ -values of the  $F$  and Hausman tests. The random effects one is the best-fitting model in the four regressions of interest.

In Table 2, they present the regression estimation results for the  $ROE_{i,t}$  model. An interesting finding is that the combined ESG or pillar-specific scores have no statistical relationship with  $ROE_{i,t}$ . This non-significant relationship could be against some previous works, such as the ones cited herein. This paper's evidence does not suggest a positive and

significant relationship between company profits and ESG scores. This result indicates no gain or a loss in earnings if the company invests in HPWP.

Despite this, regression (16) shows a potentially significant relationship between the workforce score (the quality of the HPWP implemented in the company) and the company’s profitability. Once we estimated the regression that isolates the relation of the workforce happiness or workforce score, the authors found that this relationship did not hold.

With this result, the first hypothesis, “The higher the quality of HPWP, proxied with the workforce ESG score, the higher the profitability (ROE) in a given company,” does not hold for the specific set of companies tested herein. Despite this, there is no evidence of a significant negative relationship. Therefore, European Public companies could engage in such practice without losing profits.

4.2 H2: “The higher the quality of HPWP, proxied with the workforce score, the lower the company’s market risk exposure, proxied with the CAPM beta.”  
At this point, the authors want to present the results of the fourth hypothesis test. Following the exposition of the previous hypothesis, Table 3 shows the *F* and Hausman tests of the four regressions (14–17)(14–16).

**Table 1.**  
*F* and Hausman tests  
for the ROE panel data  
regressions

| Regression number | <i>F</i> test | Hausman test |
|-------------------|---------------|--------------|
| (14)              | 0.0000        | 0.6760       |
| (15)              | 0.0000        | 0.6200       |
| (16)              | 0.0000        | 0.5020       |
| (17)              | 0.0000        | 0.7120       |

**Source(s):** Own elaboration with data from Refinitiv

**Table 2.**  
The regressions testing  
the ROE-workforce  
relationship

|                                | Dependent variable<br><i>ROE<sub>i,t</sub></i> |             |             |             |
|--------------------------------|------------------------------------------------|-------------|-------------|-------------|
|                                | (14)                                           | (15)        | (16)        | (17)        |
| <i>ESG<sub>t</sub></i>         | 0.0008                                         |             |             |             |
| <i>ESG<sub>t-1</sub></i>       | −0.0511                                        |             |             |             |
| <i>ENV<sub>t</sub></i>         |                                                | −0.0513     | −0.0992     |             |
| <i>ENV<sub>t-1</sub></i>       |                                                | −0.0149     | −0.0289     |             |
| <i>SOC<sub>t</sub></i>         |                                                | −0.1266     |             |             |
| <i>SOC<sub>t-1</sub></i>       |                                                | 0.0970      |             |             |
| <i>GOV<sub>t</sub></i>         |                                                | 0.0955      | 0.0837      |             |
| <i>GOV<sub>t-1</sub></i>       |                                                | −0.0493     | −0.0543     |             |
| <i>WorkHap<sub>t</sub></i>     |                                                |             | −0.0155     | −0.0470     |
| <i>WorkHap<sub>t-1</sub></i>   |                                                |             | 0.1125**    | 0.0695      |
| <i>α</i>                       | 17.9081***                                     | 18.0445***  | 13.4020***  | 13.6926***  |
| LLF                            | −51727.1926                                    | −51726.4521 | −51725.4481 | −51727.0609 |
| Akaike                         | 122098.3853                                    | 122088.9041 | 122086.8962 | 122098.1218 |
| Observations                   | 9,325                                          | 9,325       | 9,325       | 9,325       |
| <i>R</i> <sup>2</sup>          | 0.0005                                         | 0.0011      | 0.0013      | 0.0004      |
| Adjusted <i>R</i> <sup>2</sup> | 0.0003                                         | 0.0005      | 0.0007      | 0.0002      |
| <i>F</i> statistic             | 0.9494                                         | 6.9580      | 8.8282      | 0.9678      |

**Note(s):** Significance \* *p*-value < 0.10, \*\* *p*-value < 0.05, \*\*\* *p*-value < 0.01  
**Source(s):** Own elaboration with data from Refinitiv

As noted, the best regression estimation model in (14–16) is the random effects, and for (16), the fixed-effects one. Table 4 gives detail of each model. Following the same model sequence as in Table 2, the ESG score, the environmental and the social pillar scores have a significant positive relationship with the market risk exposure or  $\beta$  in the CAPM model. This result could be contrary to what is theoretically expected in this paper. This result is also contrary to previous works that found a negative and significant relationship between this type of score and market risk exposure. That same result holds for the workforce happiness and well-being promotion (workforce score) but with contrary results in the model (16) and (17). Despite the significance, the magnitude of the coefficients is small (less than 0.01) in practical terms. Despite being significant, the relationship between the workforce and ESG scores is still tiny to consider it strong or of practical use.

With this result, the authors conclude that the influence of the workforce in the perceived market risk exposure does not exist. Therefore, the second hypothesis does not hold for the case of the European companies tested herein.

This result suggests that the validity of the shunned stock or the errors in expectations hypotheses (Derwall *et al.*, 2011) do not hold in the European companies tested herein, and, possibly, the ESG and workforce scores should be tested with other market factor models.

| Regression number | F test | Hausman test |
|-------------------|--------|--------------|
| (14)              | 0.0000 | 0.0000       |
| (15)              | 0.0000 | 0.0000       |
| (16)              | 0.0000 | 0.0000       |
| (17)              | 0.0000 | 0.7900       |

Source(s): Own elaboration with data from Refinitiv

**Table 3.**  
F and Hausman tests  
for the CAPM beta  
panel data regressions

|                 | Dependent variable                       |                               |                               |            |
|-----------------|------------------------------------------|-------------------------------|-------------------------------|------------|
|                 | Market risk exposure or beta ( $\beta$ ) |                               |                               |            |
|                 | (1)                                      | (2)                           | (3)                           | (4)        |
| $ESG_t$         | 0.0039***                                |                               |                               |            |
| $ESG_{t-1}$     | 0.0042***                                |                               |                               |            |
| $ENV_t$         |                                          | 0.0012**                      | 0.0019***                     |            |
| $ENV_{t-1}$     |                                          | 0.0011**                      | 0.0017***                     |            |
| $SOC_t$         |                                          | 0.0006                        |                               |            |
| $SOC_{t-1}$     |                                          | 0.0006                        |                               |            |
| $GOV_t$         |                                          | 0.0023***                     | 0.0025***                     |            |
| $GOV_{t-1}$     |                                          | 0.0024***                     | 0.0026***                     |            |
| $WorkHap_t$     |                                          |                               | -0.0015***                    | -0.0004    |
| $WorkHap_{t-1}$ |                                          |                               | -0.0007*                      | 0.0009**   |
| $\alpha$        |                                          |                               |                               | 0.9674***  |
| LLF             | -395.8693                                | -375.3842                     | -367.3667                     | -1365.2634 |
| AIC             | 11235.7386                               | 11186.7684                    | 11170.7334                    | 16044.5267 |
| Observations    | 6,660                                    | 6,660                         | 6,660                         | 6,660      |
| $R^2$           | 0.0401                                   | 0.0460                        | 0.0483                        | 0.0637     |
| Adjusted $R^2$  | -0.2240                                  | -0.2174                       | -0.2145                       | 0.0634     |
| F statistic     | 109.1814***<br>(df = 2; 5,222)           | 41.9567***<br>(df = 6; 5,218) | 44.1542***<br>(df = 6; 5,218) | 4.6400*    |

Note(s): Significance: \*  $p$ -value < 0.10, \*\*  $p$ -value < 0.05, \*\*\*  $p$ -value < 0.01

Source(s): Own elaboration with data from Refinitiv

**Table 4.**  
The regressions testing  
the systematic or  
market risk (beta)-  
workforce relationship

Despite this, and similar to the previous hypothesis, the authors found no evidence suggesting that the company’s market risk increases when it engages in HPWP. Therefore, European companies could promote HPWP without increasing their market risk perception.

4.3 *H3*: “The higher the quality of HPWP, proxied with the workforce score, the better the company’s stock price percentage variation”

With the results in Table 5, the authors found that the fixed effects model is more suitable for all the regressions (13) to (15) and the random effects one for (16). Table 6 summarizes the results of the regressions. As noted, the lagged values of the social and governance pillars have a significant negative relationship with percentage price. This result suggests and confirms other theoretical positions: ESG (high HPWP quality) stocks incur higher workforce implementation costs, and, as a result, the investors tend to price that impact (Derwall *et al.*, 2011).

The same significant negative relationship holds with the previous year’s workforce (HPWP) score. The higher the score, the lower the mean or expected return in that stock. This result rejects the third hypothesis, in which the authors expected to see either a positive or a neutral (non-significant) regression coefficient.

**Table 5.**  
F and Hausman tests  
for the percentage  
stock price variation  
regressions

| Regression number | F test | Hausman test |
|-------------------|--------|--------------|
| (14)              | 0.0000 | 0.0000       |
| (15)              | 0.0000 | 0.0000       |
| (16)              | 0.0000 | 0.0000       |
| (17)              | 0.0000 | 0.0300       |

**Source(s):** Own elaboration with data from Refinitiv

**Table 6.**  
The regressions testing  
the stock price  
percentage change-  
workforce relationship

|                 | Dependent variable<br>Percentage price change |                 |                 |                 |
|-----------------|-----------------------------------------------|-----------------|-----------------|-----------------|
|                 | (1)                                           | (2)             | (3)             | (4)             |
| $ESG_t$         | -0.0320                                       |                 |                 |                 |
| $ESG_{t-1}$     | 0.0516                                        |                 |                 |                 |
| $ENV_t$         |                                               | 0.0996*         | 0.1161**        |                 |
| $ENV_{t-1}$     |                                               | -0.0013         | 0.0023          |                 |
| $SOC_t$         |                                               | 0.1174**        |                 |                 |
| $SOC_{t-1}$     |                                               | -0.1303***      |                 |                 |
| $GOV_t$         |                                               | -0.1642***      | -0.1631***      |                 |
| $GOV_{t-1}$     |                                               | 0.1389***       | 0.1391***       |                 |
| $WorkHap_t$     |                                               |                 | 0.0917*         | 0.1062**        |
| $WorkHap_{t-1}$ |                                               |                 | -0.1481***      | -0.1192**       |
| LLF             | -45912.0165                                   | -45894.4634     | -45892.3575     | -45908.2674     |
| AIC             | 107550.033                                    | 107506.9268     | 107502.715      | 107542.5347     |
| Observations    | 9,297                                         | 9,297           | 9,297           | 9,297           |
| $R^2$           | 0.0001                                        | 0.0038          | 0.0043          | 0.0009          |
| Adjusted $R^2$  | -0.1822                                       | -0.1783         | -0.1778         | -0.1812         |
| F statistic     | 0.2884                                        | 5.0518***       | 5.6476***       | 3.4608**        |
|                 | (df = 2; 7,863)                               | (df = 6; 7,859) | (df = 6; 7,859) | (df = 2; 7,863) |

**Note(s):** Significance: \*  $p$ -value < 0.10, \*\*  $p$ -value < 0.05, \*\*\*  $p$ -value < 0.01  
**Source(s):** Own elaboration with data from Refinitiv

Investing in companies that promote workforce happiness and well-being through HWPW (investing in companies with a high workforce score) could lead to lower performance against a market portfolio. Given this result (only in this test's particular perspective), investors would invest in high workforce score companies only because they want to promote the workforce's happiness and well-being practice sacrificing returns. This conclusion could lead to the belief that investing in companies with higher HPWP quality (the main interest in this paper) or higher ESG quality could lead to lower performance in the portfolio and support the shunned stock hypothesis (Derwall *et al.*, 2011).

To check the validity of this result, the authors simulated the performance of a hypothetical portfolio that invests in European companies with a high workforce score. That is a portfolio that invests mainly in companies that promote HPWP.

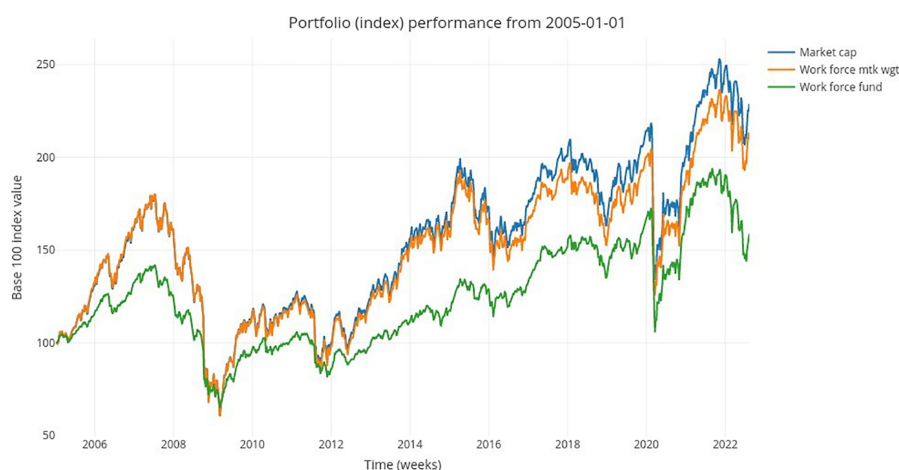
The authors made this simulation because the relationship between workforce score with profitability, market risk and stock price percentage variation modeled with the set of four regressions is a general European model. That means this relationship holds only in a yearly and aggregated perspective of the 1,537 companies included in the tests.

This general relationship, especially in the stock price percentage return, is not necessarily genuine when investing in a portfolio with companies of workforce happiness and well-being standards. For this reason, the authors simulated this hypothetical portfolio and tested the fourth working hypothesis.

*4.4 H4: "The performance of a portfolio invested in HWPW (high workforce score) stocks is better or, at least, equal to that of a market diversified portfolio"*

Figure 2 shows the historical performance of the three simulated portfolios. As noted, the fundamental or workforce-weighted portfolio (FundamentalWorkForce) is the worst-performing of these.

Following the performance summary in Table 7, the European market portfolio (MktPortfolio) paid an accumulated return of 128.76%, and the market-weighted workforce score-capped portfolio (MktWorkForceCap) had an accumulated return of 113.58%.



**Figure 2.**  
The historical  
performance of the  
simulated portfolios

**Source(s):** Own elaboration with data from Refinitiv

**Table 7.**  
The historical  
performance of the  
simulated portfolios

In [Figure 3](#), the authors present the box-plot of the portfolio returns. This figure, a one-way ANOVA test (with a  $p$ -value of 0.8540), and a non-parametric Kruskal–Wallis test of these ( $p$ -value of 1) confirm that the three simulated portfolios (especially the MktPortfolio and the MktWokForceCap portfolios) have a statistically equal return during the simulation. From the three simulated portfolios, the MktPortfolio and the MktWorkforceCap had a similar mean return (percentage variation) with values of 7.27% and 6.96%, respectively.

The level of yearly risk exposure in the simulated portfolios summarized in [Table 7](#) is 20.71%, 21.07% and 27.21% for the MktPortfolio, MktWorkforceCap and FundamentalWorkForce portfolios. Related to this risk exposure, the authors found a Sharpe ratio average value of 0.0247, 0.0213 and 0.0033 in these three portfolios, respectively. This result means that, on average, the MktPortfolio and the MktWorkforceCap earn 0.0247% and 0.0213% of risk premium, given a marginal risk exposure of 1.00%. [Figure 4](#) summarizes the historical Sharpe ratios. As noted with this visual representation and a 99.70%  $p$ -value in a one-way ANOVA test of these three-time series, the authors found that the performance of the simulated portfolios is statistically equal.

Finally, [Table 7](#) shows the CAPM-related Treynor ratio and Jensen’s alpha. From a systematic risk perspective, there is no significant extra value creation (alpha) of the two workforce simulated portfolios.

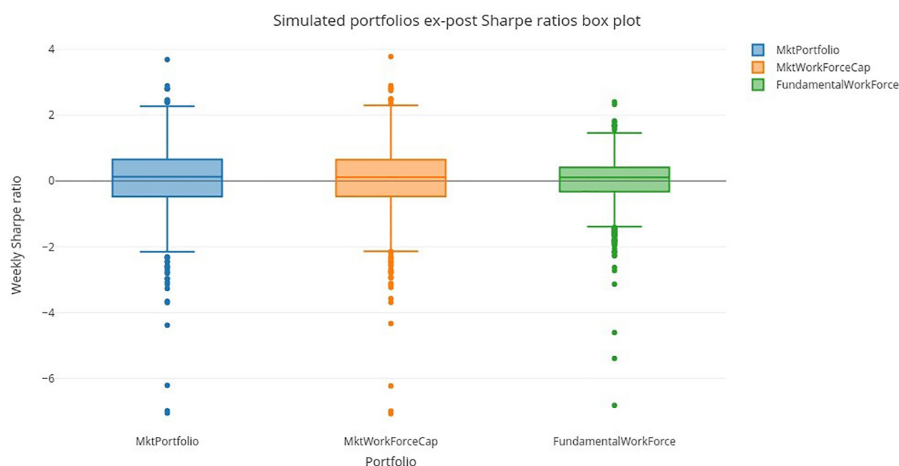
|                               | MktPortfolio | MktWorkForceCap | FundamentalWorkForce |
|-------------------------------|--------------|-----------------|----------------------|
| <i>Accumulated return (%)</i> | 128.76010    | 113.58574       | 58.71895             |
| <i>Mean yearly return (%)</i> | 7.278374     | 6.964257        | 3.836951             |
| <i>Mean yearly risk (%)</i>   | 20.71577     | 21.07116        | 27.2131              |
| <i>Mean Sharpe ratio</i>      | 0.0247       | 0.0213          | 0.0033               |
| <i>Mean Treynor ratio</i>     | –            | 0.0007          | 0.0002               |
| <i>Mean Jensen Alpha</i>      | –            | –0.0000         | –0.0000              |
| <i>Mean H-H ratio</i>         | 0.0312       | 0.0354          | 0.0000               |
| <i>Mean CvaR (95% conf.)</i>  | –0.0576      | –0.0587         | –0.0551              |

**Source(s):** Own elaboration with data from Refinitiv



**Figure 3.**  
The returns’ box-plot  
of the three simulated  
portfolios

**Source(s):** Own elaboration with data from Refinitiv



Investors of  
promoting  
happiness  
through HPWP

**Figure 4.**  
The Sharpe ratio'  
box-plot of the three  
simulated portfolios

**Source(s):** Own elaboration with data from Refinitiv

Even if it is expected that the workforce happiness and well-being promoting portfolios could overperform the market portfolio, there is no return loss by investing mainly in companies with high HPWP standards. Also, there is no difference in terms of total risk exposure against the level of the market portfolio. Therefore, the hypothesis is accepted as a summary for the test of this last fourth hypothesis.

#### 4.5 Corollary of results

To summarize the test of the four hypotheses of interest, the authors summarize their findings:

- (1) The hypothesis “The higher the quality of HPWP, proxied with the workforce score, the higher the profitability (ROE) in a given company” does not hold for the specific set of companies tested herein. There is no evidence that company profits increase if it promotes workforce happiness and well-being through HPWP. Also, no proof suggests that any of the European companies tested herein loses gains with this practice.
- (2) The statement “The higher the quality of HPWP, proxied with the workforce score, the lower the company’s market risk exposure, proxied with the CAPM beta,” does not hold for the companies of interest herein. Even though there is a positive and significant relationship between promoting HPWP and the company’s beta or market risk level, the regression coefficient levels are too small to consider the existence of a real connection between these two variables. Therefore, there is no evidence that neither suggests that the companies that promote workers’ well-being through HPWP have lower market risk nor a higher one. A company could promote HPWP without increasing its market risk perception from investors.
- (3) The results suggest that there is neither a positive nor a negative relationship. The theoretical expectation “The higher the quality of HPWP, proxied with the workforce score, the better the company’s stock price percentage variation” does not hold. Even if there is no significant relationship between the practice of HPWP and investors’ stock demand, there is also no meaningful negative relationship. Because no such

relationships hold, the authors conclude that an investor could screen her or his portfolio with mainly happiness-promoting companies without expecting lower returns due to HPWP implementation costs. This result strengthens with the HPWP portfolios simulations and the following hypothesis.

- (4) The hypothesis “The performance of a portfolio invested mainly in HWPW (high workforce score) stocks is better or, at least, equal to that of a market diversified portfolio” holds. More specifically, it fits because the performance of a market-value HPWP cap-weighted portfolio has a statistically equal performance against the market portfolio. Practitioners in the investment industry could replicate this portfolio’s weighting method to promote happiness and well-being in the workforce of companies with high HPWP without performance loss against the market portfolio. This result is especially interesting for Institutional investors (such as banks, pension funds, insurance companies, or sovereign wealth funds) who want to promote ESG and well-being practices with their investments.

## 5. Conclusions

This paper presents the company (microeconomic) and investors’ (macroeconomic) benefits of promoting the workforce’s happiness and well-being. The authors address the following question: Is there a benefit for a European company if it implements high-performing work practices (HPWP) to enhance its productivity and stock market price performance? Is there a benefit for investors if they invest mainly in a portfolio of companies primarily screened with companies with HPWP standards (workforce score)? They answered this question for the 1,983 European companies traded in the main east, west, north and south European stock markets. More specifically, companies whose stocks were members of the Refinitiv Europe price return index.

This paper assumed as true the following rationale: if a company promotes happiness (through HPWP) and well-being among its workers, they will be happier and have higher well-being (in physical, psychological and social terms). If workers feel happier and with a better well-being perception, they will be more productive. If this rationale holds, investors will benefit by screening their portfolios with HPWP stocks. With higher production, investors will demand more of the stocks of these companies, and these will have higher profits, better stock price variation and lower systematic or market risk.

To proxy the quality of the HPWP, the authors used the Refinitiv workforce score, a score that summarizes the workforce items. These score items also help to estimate the social pillar and the company’s ESG score by Refinitiv.

This paper measured the relationship between the lagged (previous year) workforce score with the ROE of a given year, the relationship of the workforce score (WFS) with the company’s market risk (beta in a capital asset pricing model -CAPM-context) and the relationship of this score with the percentage price variation of a company in a given year. The authors tested the benefits of microeconomic (WFS-ROE) and macroeconomic (WFS-market risk, WFS-stock price return).

Also, to test the benefits of promoting HPWP by investors, they simulated the performance of these 1,983 European companies’ portfolios in three scenarios:

- (1) A theoretical market-value weighted portfolio (the market portfolio or benchmark),
- (2) A theoretical market-value weighted workforce score-capped one, and
- (3) A workforce-weighted one.

The results suggest that there is no benefit in terms of profitability, market risk and stock price variation if companies promote their workforce’s happiness and well-being through

HPWP. Despite this, they also found no evidence that suggests that this practice is bad for European companies. Therefore, the European companies of the countries tested herein could enhance their workforce happiness and well-being (implementing HPWP) without losing profits, having a higher market risk, or having a worst stock price performance.

Portfolio simulations suggest no overperformance or a negative impact if investors invest mainly in stocks with high workforce well-being and happiness standards. That is, there is nothing to lose if investors promote happiness and well-being in the 1,983 European companies of interest.

This paper's results contribute to the current literature on ESG investing and workforce happiness promotion. From the company and investor's perspective, it also enhances the results of labor economics and human resources theories: Promoting happiness and well-being in the workforce does not positively impact but is not bad for the company and investor's performance.

As a potential benefit of the author's results, it is desirable that the European union labor authorities, along with the United Nations and the leading financial regulatory authorities, could promote policies to enhance the investment only in companies with high well-being standards in their workforce. The present paper gives proof to investors and companies alike to strengthen the promotion of HPWP without the loss of mean-variance efficiency or company profitability.

As guidelines for further research, the authors suggest extending the panel regressions with industry and country-specific effects. Also, they recommend extending the portfolio simulations to other regions, a country-specific perspective and an industry-specific diversification. Questions such as do these results hold in different states (regimes) of financial markets? It is also of interest. The authors believe an additional perspective that could complement the present results would be extending the current tests to a well-being pillar-specific analysis. That is, to estimate happiness, health and social relations (pillars of the workforce's well-being according to Grant *et al.* (2007) and Misselbrook (2014)) and used these in a similar pool of tests to the ones of this paper. Finally, the authors suggest repeating this research with other ESG methodologies for practical and theoretical robustness tests.

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### A1. Appendix

#### The Refinitiv ESG methodology

The authors present this appendix briefly explaining the ESG methodology and highlighting how Refinitiv calculates the workforce score. It is important to remember that the workforce score is, in the context of this research, the central approximation of the HPWP quality in a given company.

The ESG acronym refers to the three dimensions of interest in corporate social responsibility (CSR) practices: Environmental, Social, and Governance. As mentioned in the introduction, ESG is a theoretical framework that helps measure a company's CSR quality. The United Nations (UN) and some leading financial industries of the United States and Europe developed this scoring framework (United Nations, 2005, 2006; United Nations Environment Programme, 2004). Due to the UN millennium declaration and the combined global efforts of the leading financial institutions, the ESG scoring industry emerged to grade the quality of CSR practices.

Even if several publicly traded companies worldwide disclose their CSR efforts, several third-party companies research these CSR activities in companies worldwide. Among the most used in the Financial markets industry, there are Refinitiv, Robeco Sam, S&P, MSCI, and Sustainalytics. From these companies, the authors used Refinitiv's (formerly the Thomson-Reuters financial markets data branch) ESG methodology (Refinitiv, 2019). The authors decided on Refinitiv because it is the ESG-scoring company with the broadest coverage. Even if Cubas-Dias and Martínez-Sedano (2018) or Escrig-Olmedo *et al.* (2019) suggest that Refinitiv does have not the highest ESG philosophy representativeness, the broad coverage and the ESG scoring quality is appropriate for the financial industry.

Refinitiv grades public companies by measuring nearly 400 items from the company's financial reports, CSR reports, and website. Also, Refinitiv uses data from the web, social media, news companies, or even non-governmental organizations (ONG). Ten dimensions, organized in the three pillars, summarize these nearly 400 items as follows:

- (1) Environmental pillar (114 items): (a) Resource use (47 items): Related to the use of natural resources or the environmental impact of the company's activities. (b) Emissions (37 items): Items that measure gas emissions, water pollution, waste management, and related. (c) Innovation (30 items): Refers to the environmental innovation efforts in their production line, the impact of their products, or their management practices.
- (2) Social pillar (115 items): (a) Workforce quality (64 items): Measuring several practices and the existence of health, social, or psychological well-being policies, including the inclusion and diversity philosophy. (b) Human rights (14): Measuring the efforts of the company to respect human rights inside its structure or in its stakeholders. (c) Community (37 items): Refers to proper conduct and relationship with the community (or communities) in which the company trades. (d) Product responsibility (54 items): These items measure the company's product or services quality regarding their social, environmental, or health impact.
- (3) Governance pillar (123 items): (a) Management (64 items): Refer to the management and company's board practices and structure. (b) Shareholders (48 items): These items grade and

study the items related to shareholder relationships and rights (including the prevention of agency risks). (c) CSR strategy (11 items): These items summarize the quality of the ESG practices developed by the company.

- (4) Controversies (25 items): These items measure the presence of controversies, harmful practices, or actions in the resource use, workforce, human rights, community, product responsibility, management, or shareholders dimensions.

Refinitiv measures (yearly) each item in a binary, percentage, or unit-specific scale. Then scores their scale on a 0 to 1 scale and normalized the sum of items' scores with those of similar industry companies (Thomson-Reuters, 2012). With this normalization method, the industry-specific ESG leading company has a score of 100, and the worst ESG-performing one has an ESG score near zero. This method compares the CSR quality of (as an example) an Italian mining company with a US or Chinese counterpart in the same industry.

For the purposes and methodology in this paper, it is essential to mention that Refinitiv measures the 0–100 score for the whole ESG (CSR) practices, estimates the pillar-specific score, the controversies one, and the ten dimension-specific ones. From these ten dimension-specific scores, the authors are interested in the workforce quality score (dimension 2a in the social pillar), a score that grades the quality of the HPWP of a company compared with its industry peers worldwide. This workforce quality score allows the authors to proxy the promotion of happiness and well-being in a given company.

To conclude this appendix, the workforce quality score measures the following 64 items (For further details, please refer to (Refinitiv, 2019)):

- (1) Does the company have a health safety policy?
- (2) Does the company have an employee safety policy?
- (3) Does the company have a supply chain safety policy?
- (4) Does the company have an employee training development policy?
- (5) Does the company have a skills training program?
- (6) Does the company have a career development policy?
- (7) Does the company have a diversity and opportunity policy?
- (8) Does the company have diversity and opportunity targets?
- (9) Does the company have a health and safety team?
- (10) Does the company provide health and safety training to its workforce?
- (11) Does the company provide health and safety training to its supply chain?
- (12) Does the company improve the health and safety of its supply chain?
- (13) Does the company use the OHSAS 18001 safety norm?
- (14) Level of employee satisfaction (%)
- (15) Are there salary gaps in the company's workforce?
- (16) What are the average paid salaries and wages disclosed in the CSR report?
- (17) How much is the Net employment creation (in thousands)?
- (18) What is the number of employees disclosed in CSR reports?
- (19) What is the percentage of workers represented by a union?
- (20) What is the observed workforce turnover in the Company?
- (21) What is the percentage of announced layoffs to total employees in the company?
- (22) The number of layoffs announced by the company.
- (23) Did the company suffer voluntary management departures?

- (24) Did the company suffer strikes?
- (25) What is the percentage of women employees to the total workers in the company?
- (26) What is the percentage of new women employees from the new hirings?
- (27) What is the percentage of women managers from the managerial group?
- (28) What is the level of the Human Rights Campaign equality index?
- (29) Does the company use a flexible working hours scheme?
- (30) Does the company sponsor daycare services?
- (31) What is the percentage of workers with disabilities in the total workforce?
- (32) What is the number of health and safety training hours per year?
- (33) What is the number of injuries per million hours?
- (34) What is the total injury percentage (employees with injuries) in the entire workforce and contractors?
- (35) What is the total injury percentage (employees with injuries) in the entire workforce?
- (36) What is the contractors' total injury percentage (contractors' staff with injuries related to the company activities)?
- (37) What is the total number of accidents in the company?
- (38) What is the total number of contractor accidents in the company?
- (39) What is the total number of employee accidents in the company?
- (40) What is the number of occupational diseases in the company?
- (41) What is the number of employee fatalities in the company?
- (42) What is the number of contractor fatalities in the company?
- (43) What is the percentage of lost days to total working days?
- (44) What is the total production time lost due to injuries?
- (45) What is the total production time lost due to injuries in contractors?
- (46) What is the total production time lost due to injuries in employees?
- (47) What is the number of lost working days?
- (48) What is the number of lost working days among employees?
- (49) What is the number of lost working days for contractors?
- (50) Does the company sponsor an HIV-AIDS prevention and support program?
- (51) What are the average training hours?
- (52) What is the total number of training hours?
- (53) What is the amount of total training cost?
- (54) What is the average training cost per employee?
- (55) Does the company motivate internal promotion among its workforce?
- (56) Does the company have a management training program?
- (57) Does the company have an ESG supplier training program?
- (58) Did the company have wage working conditions controversies?

- (59) What is the number of wage working conditions controversies?
- (60) Did the company have diversity and opportunity controversies?
- (61) Did the company have employees' health and safety controversies?
- (62) What is the number of recent company wage condition controversies?
- (63) What is the number of recent diversity opportunity controversies?
- (64) What is the number of recent health and safety controversies?

Refinitiv uses these 64 items to score the working force score. A score used as a proxy of the quality of the HPWP promotion in the European companies of interest herein.

As an essential methodological note, the authors highlight that using other ESG scoring methodologies (such as Robeco Sam or MSCI) could lead to a different item selection and, potentially, to a different workforce quality score. This situation means that the tests and research made herein are sensitive to the ESG methodology. Given this result and because there is still no standard ESG score methodology in Europe (in the world), the authors assume that the Refinitiv ESG methodology is appropriate, given its worldwide coverage. Also, because Refinitiv is a financial data service used by most financial institutions and investors and following [Escrig-Olmedo \*et al.\* \(2019\)](#) conclusions related to the relative appropriateness (practical appropriateness) of the Refinitiv scoring methodology.

#### **Corresponding author**

José Álvarez-García can be contacted at: [pepealvarez@unex.es](mailto:pepealvarez@unex.es)

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